

Using AI Entities in Extended Reality to Reinvestigate Cold Cases

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ABSTRACT

Cold case investigations face difficulties because original crime scenes and witnesses are no longer accessible. Our early-stage concept explores the feasibility of reconstructing crime scenes in an extended reality environment, alongside AI-driven Non-Player Characters embodying the personas of former investigators and potential perpetrators. Our proof-of-concept prototype allows users to revisit crime scenes and speak to relevant parties, enabling them to iteratively refine hypotheses and reasoning strategies. We derive preliminary insights into how this system can support the formation of plausible leads and the integration of spatial evidence with narrative reasoning, contributing lessons learned toward future XR–AI investigative tools.

Index Terms: Extended Reality (XR), Virtual Reality (VR), Conversational Agents, Non-Player Characters (NPCs), Crime Scene Reconstruction, Decision Support, Cold Case Investigation.

1 INTRODUCTION

Cold cases typically refer to serious criminal investigations, most commonly homicides, that remain unsolved for extended periods of time. The absence of resolution not only impedes justice but also causes prolonged emotional distress for victims' families and communities. From an investigative standpoint, cold cases are particularly challenging due to the degradation or loss of physical evidence, the unavailability or fading memory of witnesses, and the difficulty of re-evaluating decisions made during earlier investigative phases [1]. As time progresses, the cognitive burden on investigators increases, as they must reconstruct complex spatial and narrative relationships from fragmented and often incomplete records.

In recent years, eXtended Reality (XR) technologies have been explored as tools to support law enforcement and forensic investigations, primarily due to their ability to reconstruct crime scenes in an immersive and spatially coherent manner. Prior work has demonstrated the use of virtual reality to recreate crime scenes for training, evidence review, and courtroom presentation, enabling investigators to better understand spatial relationships and event sequences that may be difficult to infer from photographs or written reports alone [2]. Such immersive reconstructions have been shown to improve situational awareness, recall, and collaborative reasoning, particularly in cases involving complex spatial layouts.

Separately, advances in artificial intelligence have enabled the creation of increasingly sophisticated Non-Player Characters (NPCs), particularly within games and interactive simulations. AI-driven NPCs are capable of maintaining contextual dialogue, exhibiting consistent personas, and responding dynamically to user input, thereby enhancing realism and engagement [3]. Beyond

entertainment, such agents have been explored in domains such as training, education, and role-playing simulations, where they can embody distinct perspectives or expertise and facilitate exploratory reasoning.

Despite these advances, XR-based crime scene reconstruction and AI-driven NPCs have largely evolved in isolation, and their combined potential for investigative support remains under-explored. Existing XR systems primarily function as passive visualization tools, while investigative reasoning, such as questioning witnesses, challenging assumptions, or revisiting alternative narratives, remains external to the virtual environment. This separation limits the extent to which investigators can iteratively test hypotheses and integrate spatial evidence with narrative accounts.

In this work, we present an early-stage concept that integrates XR crime scene reconstruction with persona-driven AI NPCs to support cold case investigations. Rather than proposing a finalized tool, our primary contribution lies in the system design approach and the insights gained from exploring how immersive environments and persona-driven agents can be combined to support investigative reasoning. By embedding AI NPCs that embody the viewpoints of former investigators, victims, and potential perpetrators within an XR reconstruction, our prototype allows investigators to spatially revisit past events while engaging in interactive dialogue.

We demonstrate our approach through the reconstruction of a resolved cold case, implemented as a proof-of-concept XR prototype featuring AI-generated assets and multiple AI NPCs with distinct investigative personas. Preliminary observations inform a set of lessons learned regarding persona design, interaction flow, grounding of AI responses, and ethical constraints, highlighting the exploratory potential rather than validated effectiveness of XR–AI integration as a complementary direction for cold case analysis.

2 METHODOLOGY

To evaluate the feasibility of integrating XR crime scene reconstruction with AI-driven NPCs for cold case investigations, we reconstructed a previously resolved homicide case and examined whether interaction within the system could yield investigative outcomes consistent with the known resolution. An overview of our method can be seen in Figure 1.

2.1 Case Study

The selected case is a homicide that was left opened for three years, and was resolved in 2019 despite limited physical evidence, making it a suitable benchmark for assessing investigative reasoning rather than evidence abundance. Extensive media coverage provided publicly available accounts from multiple stakeholders, enabling the reconstruction of key environments and evidence as XR assets, as well as the design of persona-driven AI NPCs representing an investigator and a potential perpetrator. System effectiveness was assessed by determining whether users interacting with the reconstructed scenes and NPCs could form plausible leads and reasoning pathways aligned with the case's established outcome.

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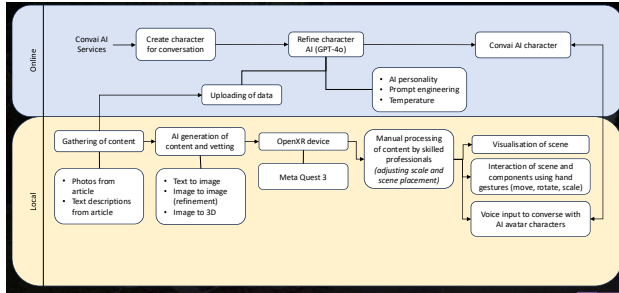


Figure 1: Method Overview

2.2 AI Generated Assets

To reconstruct the physical context of the crime, we generated three-dimensional assets representing key evidentiary items described in publicly available case materials. Given the heterogeneous nature of the source data, we employed both image-to-3D and text-to-3D generation approaches to accommodate varying levels of visual documentation.

Image-to-3D asset generation. For evidentiary items accompanied by photographs in news articles, three-dimensional assets were generated using an image-to-3D generation model (Tencent Hunyuan 2.0 3D). Photographic images of physical evidence were used as input to produce 3D objects that preserved salient visual characteristics from the source material, enabling higher-fidelity reconstruction of evidence when visual references were available.

Text-to-3D asset generation. For evidentiary items without photographic documentation, three-dimensional assets were generated using text-based prompts derived from descriptive accounts in the news articles. Prompts encoded key attributes such as object type, material, and contextual relevance, allowing representative 3D models to be created in the absence of visual input.

2.3 AI NPC Personas

To support interactive investigative reasoning, we implemented AI-driven Non-Player Characters (NPCs) representing former investigators and potential perpetrators involved in the case. Each NPC was prompted with case-specific information as well as a predefined persona reflecting their role in the investigation. For example, the perpetrator NPC was configured to exhibit a more defensive and evasive conversational style, while the investigator NPC adopted a calm and methodical demeanour.

The personas were manually defined for this exploratory prototype and do not attempt to model the actual personalities of the real individuals involved, due to both the early-stage nature of this work and the limited publicly available information. These persona constraints were used to guide response tone and behaviour during interactions, rather than to introduce new factual information beyond the documented case materials.

2.4 Interaction Flow

Users navigate the XR environment to explore reconstructed crime scenes, examine AI-generated evidence, and interact with NPCs to gather information. Movement is controlled through standard VR navigation, allowing users to approach evidence or NPCs and inspect items from multiple perspectives. While exploring, users can pose questions to NPCs via voice input, prompting responses that are both grounded in the documented case materials and guided by the NPCs' predefined persona frames. This ensures that each NPC maintains role-consistent behavior, such as a

defensive posture for a perpetrator or a calm, analytical style for an investigator, while providing relevant information to support investigative reasoning.

NPC responses are generated in real time through GPT-4o and delivered via text-to-speech, synchronized with avatar animations to maintain immersion. Users can iteratively combine information gathered from NPC interactions with spatial observations of the environment and evidence to form, test, and refine hypotheses. This iterative workflow simulates the investigative process, allowing users to reconstruct sequences of events, identify plausible leads, and evaluate potential scenarios.

3 RESULTS & CONCLUSION

Our preliminary observations suggest that the XR-AI system effectively supports investigative reasoning by enabling users to engage with reconstructed evidence and persona-driven NPCs. Interactions with NPCs often prompted users to consider alternative hypotheses, revealing that even a small set of role-consistent personas can generate intellectual friction and broaden investigative perspectives.

Several points of friction were also observed. NPCs occasionally exhibited overconfidence or drifted from their persona, and the underlying LLM sometimes produced ambiguous or inappropriate responses. Despite these limitations, the system covered a broad range of evidence items, allowing iterative exploration of the crime scene and hypothesis testing.

These behaviors highlight the potential of persona-driven NPCs in VR investigative tools: by simulating multiple viewpoints and encouraging critical questioning, they support deeper engagement with the case material and reinforce memory through spatial anchoring.

4 FUTURE WORKS

Limitations of this study include a small sample size, the fidelity of AI-generated assets, constraints in LLM accuracy, and ethical considerations when simulating sensitive roles. Future work will focus on conducting qualitative and quantitative evaluations to assess the system's effectiveness, usability, and impact on investigative reasoning. This includes user studies with domain experts and the collection of performance metrics to validate the observed benefits. Additionally, we aim to develop more accurate and grounded AI NPC representations by constructing personas directly from documented data about the real individuals involved in the case, thereby improving authenticity, consistency, and investigative relevance.

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